AN ADJOINT PAIR WITH AN EXACT LEFT ADJOINT TRANSFERS ENOUGH INJECTIVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the only step it delegates is the named same-chapter lemma carried on this card as a navigation stub, and the arbitrary object it starts from makes the conclusion exactly the statement about enough injectives.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- A functor with an exact left adjoint preserves injectives. Same chapter, Lemma lemma-adjoint-preserve-injectives. Supplies that the right adjoint carries injective objects to injective objects, which makes uI an injective object of the target category.
- Transformation of injective maps and enough injectives. Same chapter, the notions of a functor transforming injective maps into injective maps and of a category with enough injectives. Supply the hypothesis that the embedding in the source category is preserved and the definition of enough injectives that is being verified for the target.

Source reference. Original lemma 015Z: sources/stacks-homology/pinned-original-source.tex, source lines 7487–7487. Navigation pointer invoked at proof line 7537; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. Let $u : \mathcal{A} \to \mathcal{B}$ and $v : \mathcal{B} \to \mathcal{A}$ be additive functors. Assume*

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(1) *u is right adjoint to v,*
(2) *v transforms injective maps into injective maps,*
(3) *$\mathcal{A}$ has enough injectives, and*
(4) *$vB = 0$ implies $B = 0$ for any $B \in \mathrm{Ob}(\mathcal{B})$.*

Then $\mathcal{B}$ has enough injectives.

Proof. Pick $B \in \mathrm{Ob}(\mathcal{B})$. Pick an injection $vB \to I$ for I an injective object of $\mathcal{A}$. According to Lemma 015Z and the assumptions the corresponding map $B \to uI$ is the injection of B into an injective object. □